Cheat Sheets

Supporting Migrants To Make Informed Decisions When Accessing Education Abroad

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1.0 Introduction

1.1 Global Mobility as a Force for Good

The United Nations Sustainable Development Goals (UNSDG) took effect in January 2016 as a global framework for equitable growth amid the climate crisis (Sustainable Development Goals (SDGs), 2015). The 17 goals span industry, agriculture, politics, and social life, providing a blueprint for addressing global challenges with environmental concern. Our project focuses on goal 10.7, exploring how migration reinforces or alleviates global inequities.

10.7 Facilitate orderly, safe, regular and responsible migration and mobility of people, including through the implementation of planned and well-managed migration policies.

Migration can mean safety from persecution, improved education, better employment opportunities, and more robust public services. Global mobility can also benefit home countries by increasing wealth, knowledge, and investment through return migration. Migration also benefits transit and target nations by increasing economic activity and cultural diversity (IOM, 2020, p. 9). However, migration can also create opportunities for harm, exposing migrants to exploitation, abuse and social exclusion (Living in the Shadows: A primer on the human rights of migrants, 2021). Mitigating against these harms has never been more important, which means working with and for immigrant communities to support safe, structured migration that benefits all.

1.2 Our Project, In Brief

Our project aimed to tackle the issue of migration in education, working with and for international students over 13 weeks to highlight and address key barriers to accessing education abroad. As a group, we conducted face-to-face interviews with 10 diverse international students across UCD to learn from their experiences, motivations, and challenges. As a response to the 'information overload' many of our stakeholders experienced, we created a knowledge platform that supports the exploration, planning and comparison of different countries, universities, and local areas.

2.0 The Discovery Phase

2.1 Selecting our UNSDG

To launch our project, we explored the 17 UN SDGs as a group to decide which of the goals most aligned with our aims and values. Initially, we were drawn to the goals surrounding social inclusion and inequities, such as 4: Quality Education, 16: Peace and Justice, and 10: Reduced Inequalities. Given that our most accessible population was third-level students, it made sense to explore education. Considering our group mostly consists of international students, it also made sense to tackle immigration, as we could offer an insider perspective. Therefore, we decided to address Goal 10.7: Facilitate orderly, safe, regular and responsible migration, viewing the topic through the lens of both global migration and accessing education.

2.2 Why Be User Centred?

User-Centred design is a design philosophy in which end-user needs and capabilities guide design decisions across all stages, with user testing informing iterative refinement (User-Centred Design, n.d). In our project, we applied several key principles of user-centred design:

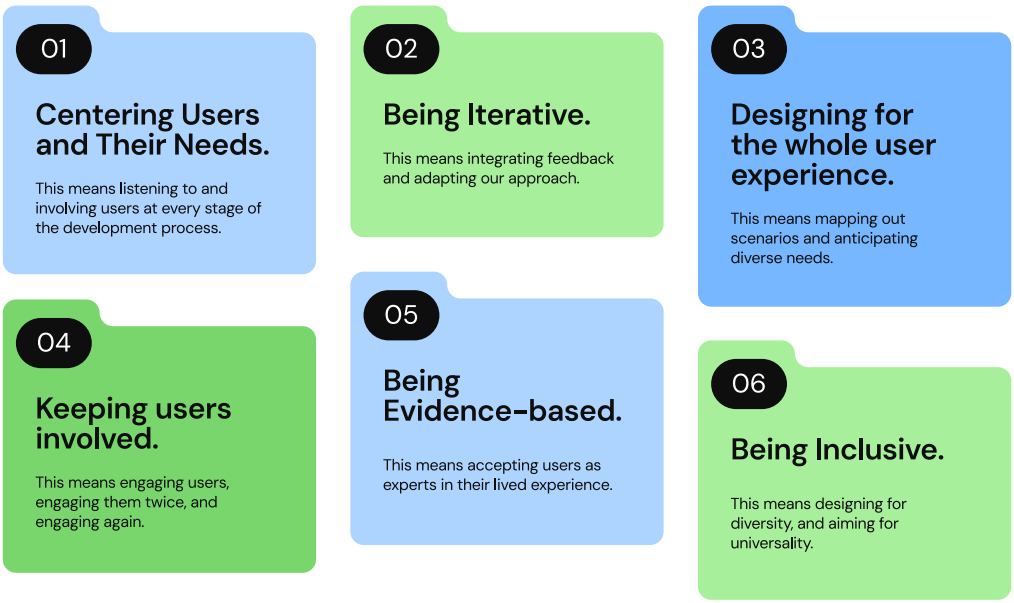


Figure 1: User Centred Principles

Regarding our project, addressing barriers to accessing education abroad, there already exists some literature which is focused on user-centred design with/for immigrants and with/for international students. One example is Virtual Volunteer, a collaboration between IBM, the International Federation of Red Cross and the Red Crescent Movement. The team interviewed Syrian migrants in Sweden alongside local volunteers to uncover challenges and perspectives from both sides, ultimately developing targeted digital prototypes for each group (Munasinghe, 2017). Onboarding without Borders focuses on the international student experience. After detailed interviews with students navigating studies abroad, the project uncovered gaps in university onboarding processes. The proposed solution was a modular, unified digital platform designed to work globally, helping students integrate more smoothly, wherever they studied (Ashoftehfard, 2025).

2.2 Immigration Journals

Before engaging users, we decided to map our own immigration experiences. This helped us identify key barriers and recurring themes to use during our interviews. First, we mapped the Irish immigration system by department, stakeholders, and user touchpoints. This helped us to structure our journals.

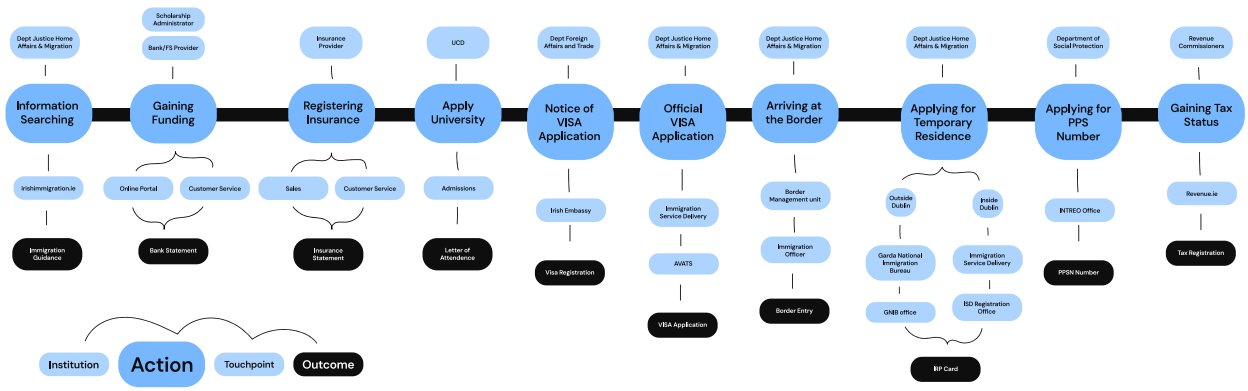


Figure 2: System Map of Irish Immigration Network

Using this system map as a guide, we then broke down our immigration experience into four key stages:

Stage 1: Researching and Planning

Stage 2: Applying to University

Stage 3: Applying for a Visa

Stage 4: Post-Arrival Tasks

Then, each group member created an immigration diary, mapping their experiences across the four themes. At this stage, we identified geopolitical concerns, safety, information accessibility, issues with immigration agencies, lack of unified information, delays, lack of cultural knowledge, housing, and costs as key barriers. After deliberation, we decided the most critical stage of the immigration process was the very beginning, as we wanted to address the barriers that might cause students to give up on their journey before it even begins.

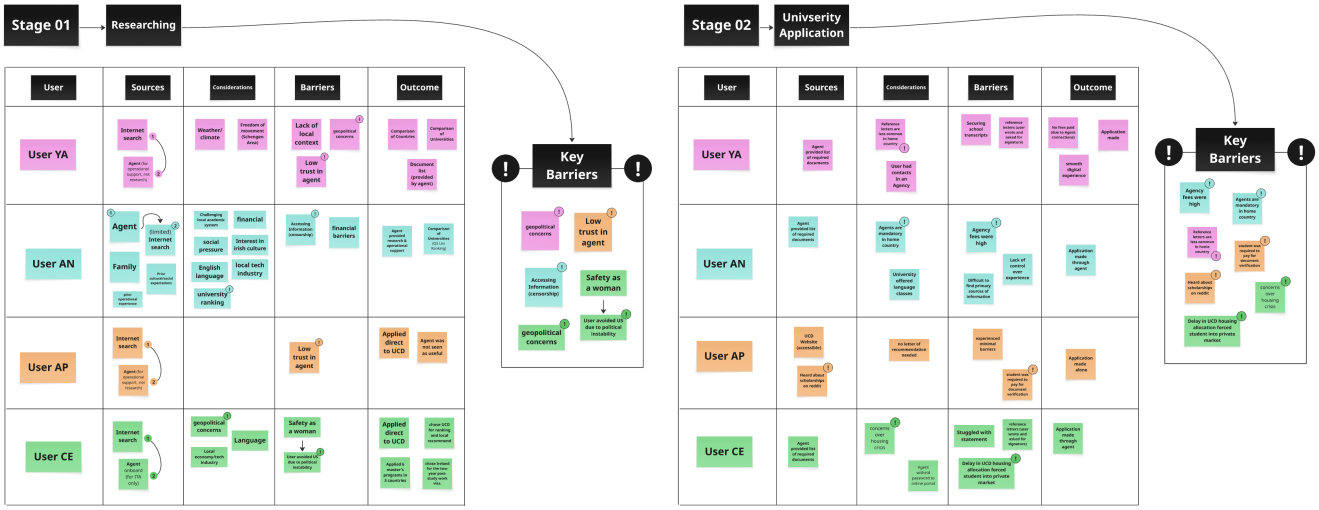


Figure 3: Immigration Journals & Barriers

3.0 Prepping For User Engagement

3.1 Establishing our Themes

Drawing from the barriers in our Immigration Journals, we divided our topics into four themes, each examining its impact on research and decision-making:

1. Local Media and Culture
2. Preparing for Life in a New Country
3. Cost of Living Concerns
4. Administrative Barriers

3.2 Writing our Interview Guidelines

Using these themes, we structured our interview guidelines, guided by extensive user engagement research. For each theme, we wrote four open-ended questions with optional scripted follow-ups. We included timing guidance, targeting ten minutes per section (see Appendix A: Interview Guide).

3.3 Writing our Mission Statement

We drafted a group mission statement at this stage, fostering alignment on aims and maintaining project focus. Our final mission statement was:

Addressing Barriers to Informed Decision Making When Accessing Education Abroad

4.0 User Engagement

4.1 Interview Process

Our team conducted interviews by reaching out to students in our networks. We contacted nineteen prospects, with nine agreeing to participate. We managed all logistics: sending structured emails, arranging time slots, and collecting consent forms before scheduling (see Appendix B: Communication Materials). Interviews were audio-recorded for accurate transcription. Recordings were kept in our project Google Drive for 24 hours before prompt deletion. Consent forms covered recording permissions and withdrawal rights at any stage.

Two team members attended each interview: one leading the discussion and the other taking notes to ensure accurate data collection and prevent the omission of insights. The interviews followed a semi-structured format and lasted approximately 30 minutes each. They focused on three broad themes: the participants' journey to Ireland, their understanding of Irish culture, and their experiences with housing and the cost of living. This ability to probe further during the session made face-to-face interviews particularly valuable. After each interview, participants completed a Google Form questionnaire (see Appendix C: Questionnaire & Answers).

4.2 Interview Findings: Isolation, Fragmentation, & Decision Paralysis

The results from our interviews were clear: many users reported confusion, frustration and decision paralysis during the discovery phase of their immigration process. One recurring theme was a lack of local context; many participants reported anxiety and apprehension about moving abroad because they lacked a clear understanding of life in their target country. The majority of our participants reported using social media platforms and AI to research countries and universities, and many expressed concerns about the reliability of the information. The second-largest source of information was personal networks, where users relied on family or friends to inform their choices.

We found that users with support networks abroad struggled far less, often selecting countries based on their recommendations and guidance. Users without a network expressed a desire for personal connection, guidance from peers, and trustworthy advice. Many users said they didn't know where to start, often using platforms like Reddit to jump-start their decision-making. When planning, many users reported scattered information as a major barrier, often finding it difficult to find a single reliable source.

Another major theme we identified was a desire to compare countries, universities, and local areas. Many of our users reported difficulty making comparisons across multiple criteria. Some used AI to make these comparisons but had doubts about the information's reliability; others simply made comparisons in their heads or discussed their options with friends or family. Excluding participants who selected UCD without considering other options, all participants found making comparisons important but challenging. Some expressed concern about missing opportunities by 'writing off' certain countries due to incomplete or biased information.

When discussing the types of information users sought, most cited concerns about the cost of living and housing. Users were also interested in the economic health of their target countries, with many seeking nations with large technology sectors. Our users also expressed interest in the global university rankings and the availability of student housing. Participants also wanted to know how friendly or walkable their city was, or what the local climate was like. Finally, many were curious about the local culture, but struggled to find useful information that didn't rely on stereotypes.

4.3 User Personas

Our goal was to create two distinct personas with distinct motivations, backgrounds, and challenges to reflect the diversity of our user base. These personas were developed using demographic data from our questionnaire, behavioural insights gathered during interviews, and our design intuition to provide realistic yet non-identifying details.

Figure 4: User Personas

Marisol Valeria Marquez

Cheat Sheets

Age: 27 | Nationality: Mexican | Income: $1200 pm

Home: Mexico City | Job: Teacher | Priority: Budget



Background

Marisol has a **bachelor's degree in Education and Child Nutrition**, she works as an assistant teacher in Mexico City for underprivileged children.

Motivations

She wants to pursue a Master's Degree in education, she dreams of working for the **European Union or the United Nations** on education reform. She wants to move to **Europe or the US** to be closer to her dream job, she also believes the quality of education will be higher.

Barriers to Education:

She is concerned about the cost of living in Europe and the US compared to Mexico, her top priority is **affordability**. She is struggling to research her options as she doesn't know where to start. She wished she had a family member abroad who could give her **guidance and support.**

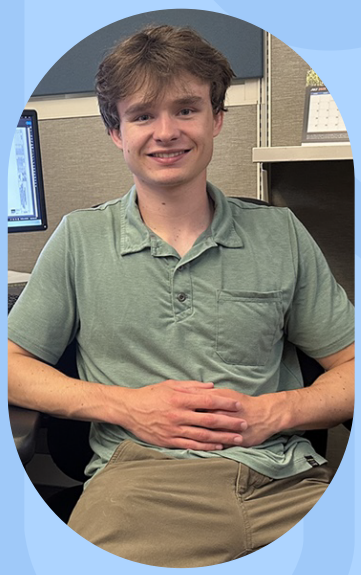
Figure 4 (a) User Persona 1: Marisol Valeria Marquez

Daniyar Khusanov

Cheat Sheets

Age: 22 | Nationality: Khazakhstani | Income: $2100 pm

Home: Astana | Job: Tech Intern | Priority: University Ranking



Background

Daniyar holds a degree in **Computer Science** and has recently completed a technology internship. He's now exploring further education to help him **navigate a difficult job market**.

Motivations

Daniyar wants to pursue a Master's Degree in Business to complement his computer science background. After graduating, he wants to work in the tech industry as a consultant or start his own business. **University ranking and reputation** is important to Daniyar as he hopes it will help him find a job. He also wants a country with a **strong economy and tech industry**, focusing on EU countries due to freedom of movement.

Barriers to Education:

Daniyar doesn't know anyone who's moved to the EU before, and he's struggling with isolation and a lack of support with administration. He has a general idea about what he's looking for, but he's **struggling to compare countries** and universities. He's eager to find support to help with his journey.

Figure 4 (b) User Persona 2: Daniyar Khusanov

5.0 Ideation

5.1 Finalising Our Requirements

Based on the findings from our user interviews, we moved forward to establish the requirements for our interface. For Functional Requirements, the ability to compare, access information on a single platform, plan, and connect with peers was most important. For Data Requirements, we found that economic data, cultural knowledge, local context, and university reputation were the most relevant. For Social Requirements, we established a need for authentic connection, collaboration, and community building when planning to emigrate. For Environmental Requirements, we decided that offline functionality, cross-platform access, and an affordable price point were key features.

5.2 Competitor Analysis

Study Portals

Study Portals is a global platform that helps students explore study programs and countries. It also offers country guides that provide useful information for travellers planning to visit a country.

Why is it a competitor?

It helps users explore countries and study options, and offers detailed, country-specific guides to help them make informed decisions. Although it's not as broad as our platform, it still deals in areas core to our idea.

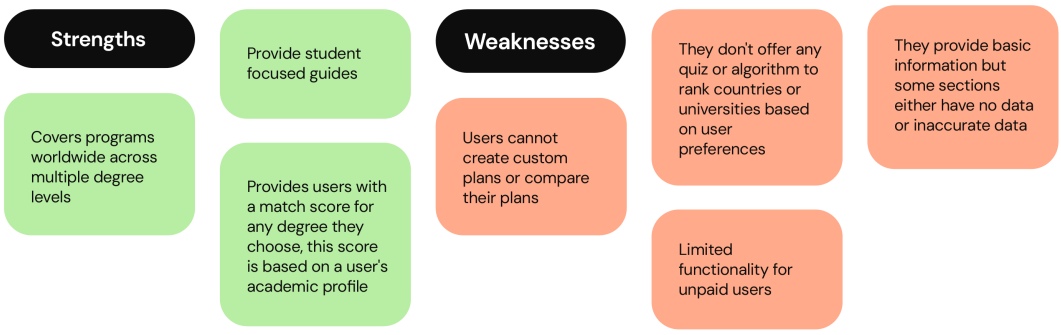


Figure 5: Study Portal Analysis

Numbeo

Numbeo is not a direct functional competitor, but is a substitute competitor. In UX and product design, this is often referred to as a "latent competitor". According to Numbeo (n.d.), it is an online platform that allows users to explore, share, and compare cost-of-living data worldwide. It offers a range of features, including:

- Viewing up-to-date price information
- Utilising the wisdom of the crowd (crowdsourcing) to ensure the highest possible accuracy
- Providing tools for conducting systematic research on the cost of living and property markets
- Calculating key indices such as the Cost-of-Living Index and Local Purchasing Power

Why is it a competitor?

It provides the core information users need when deciding where to immigrate or study. Users may rely on it as an alternative to making informed decisions.

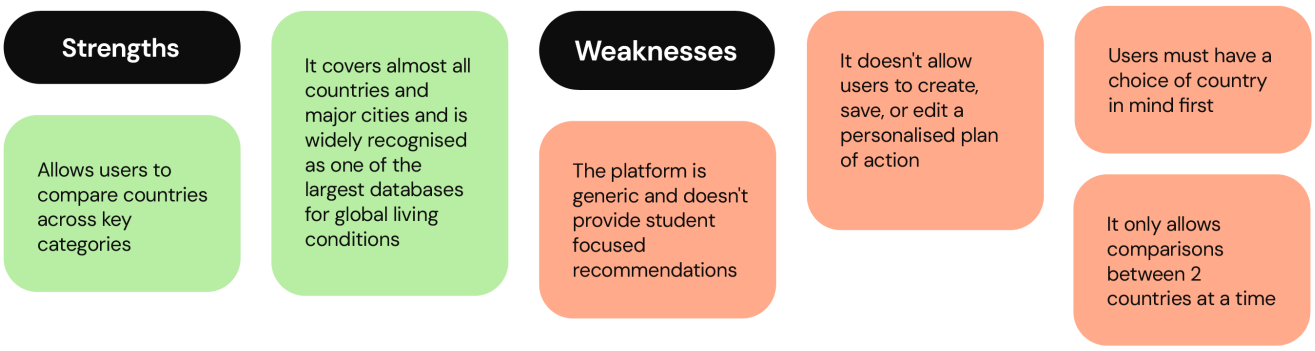


Figure 6: Numbeo Analysis

5.3 Initial Ideation

To generate our concepts, we first gathered the different requirements into themes or topics. Some of the requirements were conflicting, but others were natural complements, such as the need for collaboration and community. Once we'd divided our requirements, we ideated interfaces that could address these needs and support our wider mission. Some of our initial ideas included:

Migrant Storytelling - Inspired by Dollar Street

A content platform where users can document their lives and experiences using images, videos or text. This concept aimed to address a lack of local context and unreliable information by connecting migrants through shared storytelling.

Positivity Platform - Inspired by The Happy Newspaper

Some of the topics discussed involved delicate issues such as safety, anti-migrant sentiment, and economic injustice. This idea addressed these issues through a positive lens, only featuring countries with a high friendliness index, generosity score, or women's safety score.

Student Housing Rights

This platform addressed cost-of-living and housing issues by assessing countries using a tenancy rights index, rent as a proportion of monthly income, and the ratio of student beds to student population.

Immigration Cheat Sheets (ICS)

This platform addressed the barrier of planning, fragmented information and the need to make comparisons. This platform would give students the tools to build their own 'cheat sheets' for a certain university and city. Then it would display relevant data, such as weather, grocery costs, or walkability in a local area.

Global Study Hub (GSH)

This concept addressed the need for planning, community and collaboration, creating an administration and planning hub based around a timeline motif. Users can collaborate to navigate their immigration journey, prepare for major milestones, warn peers about deadlines, or provide emotional support.

Ultimately, we decided to move forward with Global Study Hub and Immigration Cheat Sheets, as these concepts aligned best with the requirements and had the greatest potential for further development. Full ideation artefacts, requirement notes, and early sketches are provided in **Appendix D–F**.

6.0 Immigration Cheat Sheets

6.1 Overview

ICS is a platform that cuts through the noise and supports informed planning, comparisons, and decision-making for student migrants. Users who do not know where to start can complete a quiz based on their preferences, which provides a list of recommendations for suitable countries, universities, and local areas. Users who already have an idea in mind can proceed directly to the planning phase, creating personalised immigration plans and comparing them based on criteria such as cost of living, global rankings, weather, or walkability. Users who want to explore can browse a list of pre-made cheat sheets and use filters to narrow their research.

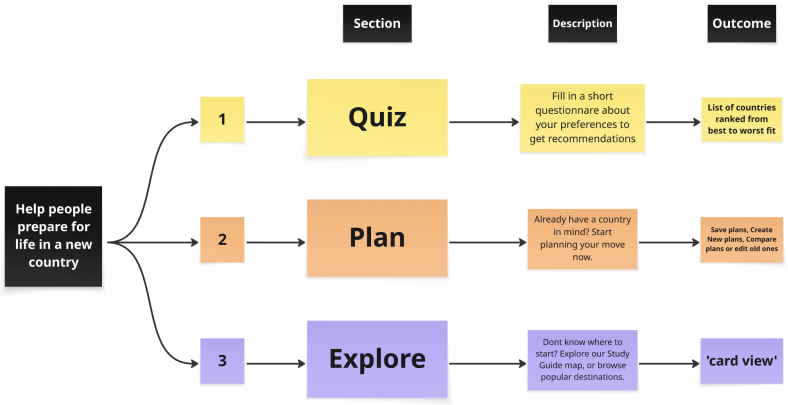


Figure 7: Initial Sitemap ICS

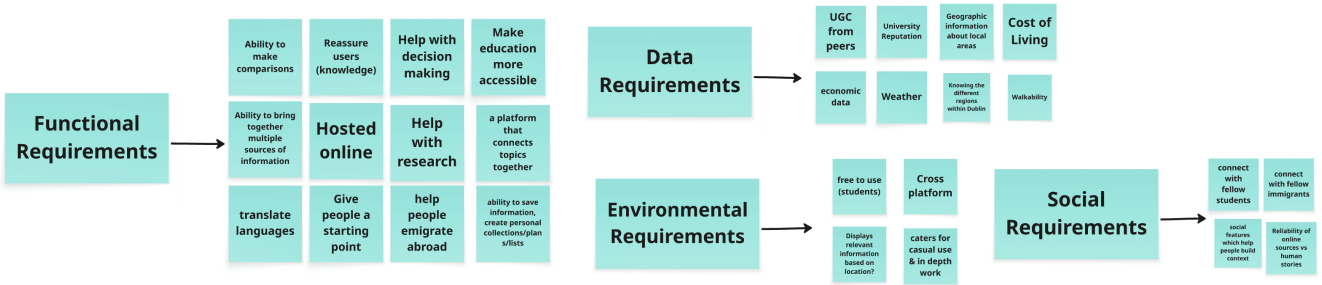


Figure 8: Mapping the Requirements for ICS

6.2 Lo-Fi Prototype

The ICS prototype consists of three main modules: Quiz, Plan, and Explore. The full Lo-Fi prototype is available at this link:

https://www.figma.com/proto/61VZ5yh9BnuHwqBuJn4yWJ/Cheat-sheet?node-id=77-432&t=LDbQcvutQ6Kowh5T-1

1. Quiz Module

We created a 7-question quiz to help users find countries and universities that best match their preferences. This section guides users through seven key questions. For questions involving continuous variables, such as budget or ranking preference, users adjust values by dragging left or right. Other questions use simple taps for selection. To minimise potential language barriers, we included supportive visual cues throughout the quiz.

7-question quiz:

1. How much is your budget for cost of living and lifestyle?
2. What is your acceptable commute time and ideal city experience?
3. What kind of climate do you prefer?
4. What languages do you plan to use?
5. What is your general field of study?
6. What university ranking do you prefer?
7. What are your long-term career goals?

2. Plan Module

The Plan section helps users build and compare migration plans. Users are encouraged to log in to save their plans. After signing up or logging in, they can begin planning by selecting a university, city, or field of study. The system then compiles the selected information into a structured list that includes multiple data points:

Country: climate, income ratio
University: tuition fees, ranking ratio
Local Area: rent, cost of living, walkability, and travel time

Users can create multiple plans and add them to the comparison area for side-by-side evaluation. Plans can also be saved to the Favourites section. If the user previously completed the Quiz, the Favourites section automatically includes the university profiles they saved from the quiz results, enabling a more complete comparison.

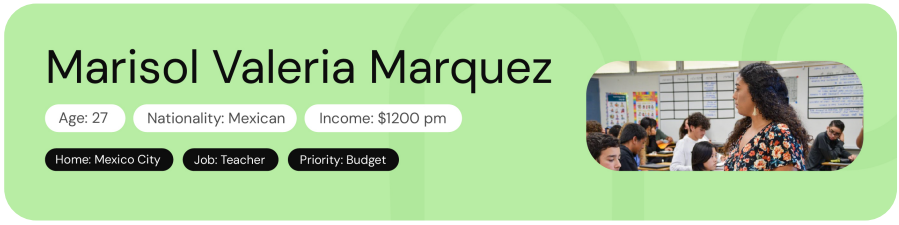
3. Explore Module

The Explore module offers two pathways for discovering content:

Categories or Feeds: Users can browse the full feed or filter by category.
Country-based exploration: Users interact with a world map, select a country, and swipe through recommended content or relevant profiles.

Both paths include interactive elements that allow users to open details, explore recommendations, or save items to Favourites.

6.3 User Scenarios



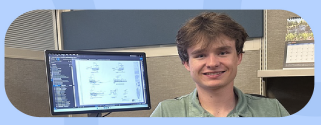
Marisol is motivated to use ICS because of its simplicity and accessibility. She hopes to make quick and easy comparisons between countries and universities, focusing on affordability. When she opens the app, she can take a quiz, create a cheat sheet, or explore user-generated sheets. Marisol already has a rough idea of what she's looking for, so she chooses the recommender quiz. She selects a comfortable lifestyle for a maximum budget of $2,500 a month. She doesn't mind university rankings but sets a maximum fee budget of $6,000. Marisol identifies walkability and a short commute as priorities, believing they will help her focus. She also applies a filter for the US and Europe, favouring a warmer climate similar to her home country.

After completing the quiz, Marisol receives a visual breakdown of her priorities, showing how closely her top recommendation aligns with her goals. Her first recommendation is San Vitale in Italy, where she attended the University of Bologna; her local area is determined by proximity and affordability. Her second recommendation is Horta-Guinardó in Spain, where she is attending the University of Barcelona. She notices her fees would be more affordable than in Italy, but her commute time would be much longer. Marisol feels empowered by her ability to easily compare the two options and decides to create an account to save them.

Daniyar Khusanov

Age: 22 | Nationality: Khazakhstani | Income: $2100 pm

Home: Astana | Job: Tech Intern | Priority: University Ranking



Daniyar is motivated to use ICS because he wants to make easy comparisons between countries and universities. When he opens the app, he's given the option to take a quiz, create a cheat sheet, or explore user-generated sheets. He already has a clear picture of his options, hoping to move to Ireland or Germany because of their strong industries, so he decides to create two cheat sheets manually.

He selects the Belfield area of Ireland and attends University College Dublin, hoping to secure student accommodation. After saving his first cheat sheet, he selects the Arcisstraße area in Munich, Germany, and chooses the Technical University of Munich, and he also adds a filter for a strong tech industry.

Daniyar can easily compare the immigration plans, noting that TUM has a much higher ranking than UCD and higher average salaries. Dublin ranks higher for a strong tech industry.

ICS makes Daniyar feel confident in his decision to choose TUM, leaving UCD as a strong second candidate. The platform helped him cut through fragmented, contradictory online information to make an informed decision about his education.

7.0 Global Study Hub

7.1 Overview

GSH is a platform that brings student migrants together to plan their immigration journey and build communities, all centred around a timeline motif. To accommodate users during the discovery and planning phases, we established two entry pathways for this interface. Users who want to explore previous immigration plans can use the search functionality to view archived plans or join active planning communities. Users who have already selected a target country and wish to plan their journey can create a hub, invite friends, and map milestones on their timeline.

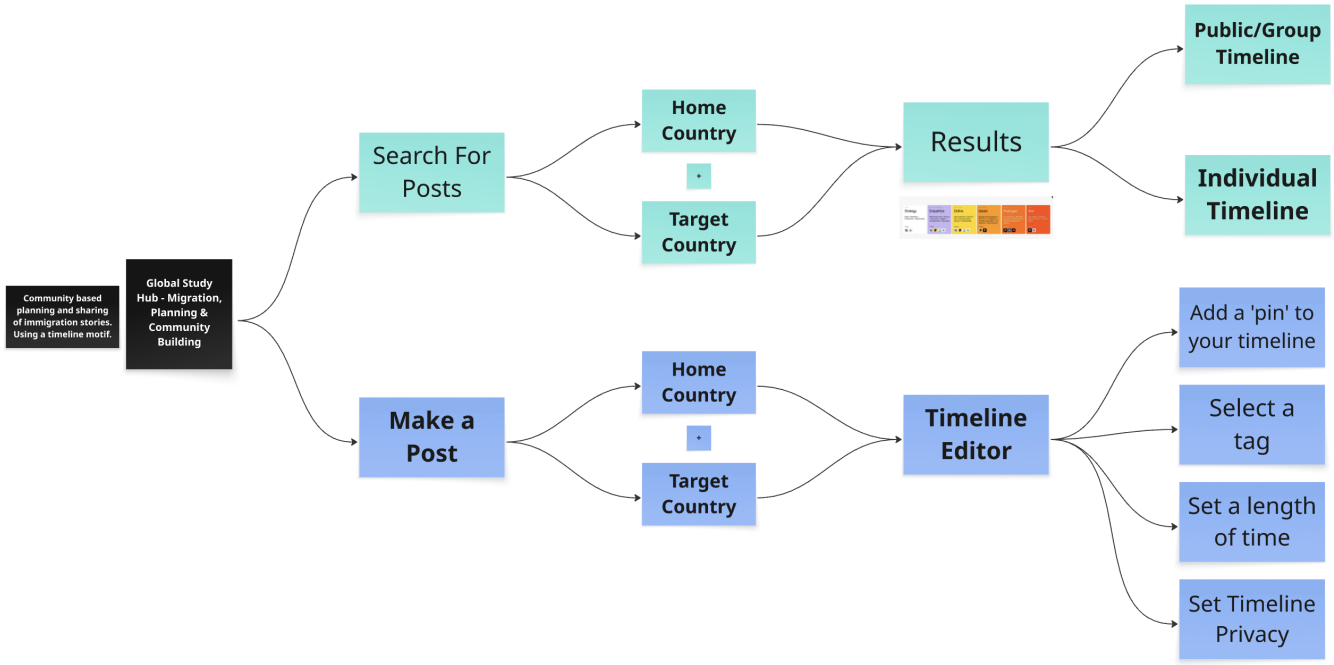


Figure 9: Initial Sitemap GSH

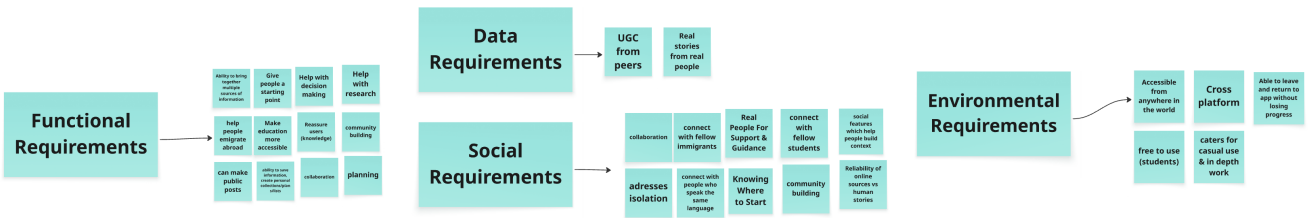


Figure 10: Mapping the Requirements for GSH

7.2 Lo-Fi Prototype

The central motif of this interface is a timeline with editable tabs and pins: a tab represents a length of time, and a pin represents a single point in time. Then they can add pins along the timeline for specific milestones, such as submitting their documents. When a user creates a pin, they can select which tab it's connected to from a drop-down list so that associated pins and tabs are the same colour. If a user invites friends or opens their hub to members, they can collaborate on their journey together by setting reminders for deadlines, linking to useful web pages, or coordinating their applications.

When a user chooses to create a tab, they must drag their finger/mouse to lengthen or shorten the tab. When a user creates a pin, it appears on the timeline and follows their mouse cursor until they click to place it. A user can tap on a pin/tab to enter ‘detailed view’, where users can change the title or add text using the editor button. When a user first creates a tab/pin, this detailed view will appear, with editable text fields, allowing them to add details and save when complete. In the detailed tab view, associated pins will also be shown in a list, aiding navigation and context. The full Lo-Fi prototype can be found here:

https://patrickfeeneydesign.com/GlobalStudyHub.pdf

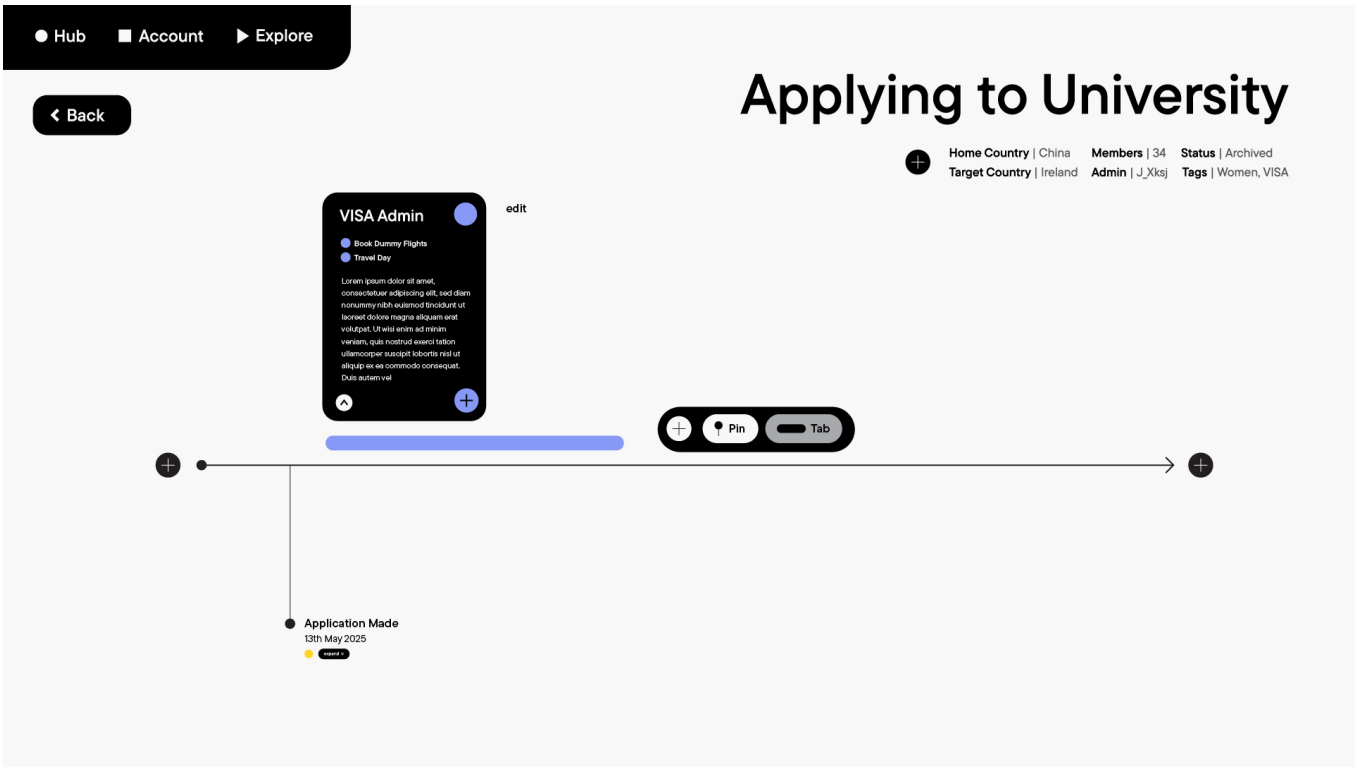
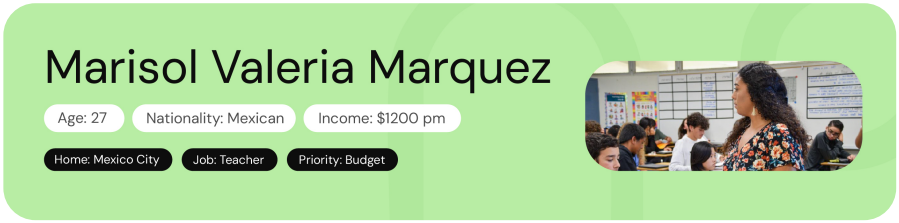


Figure 11: GSH Timeline Editor View

7.3 User Scenarios



Marisol is motivated to use GSH because of its in-depth, community-generated content, noting that she can view previously published immigration timelines. When she opens GSH, she's presented with two options: to create a hub or search for a hub. Marisol isn't at the stage to begin planning her journey quite yet, so she decides to search for a previous immigration timeline for inspiration. When searching for a hub, she first selects her home country, Mexico, then the USA and a variety of European countries as her target countries. Then she adds a filter for 'budgeting', hoping to find a study hub that focuses on costs. Marisol is presented with a variety of plans, and the search results clearly indicate the home and target countries on each hub preview.

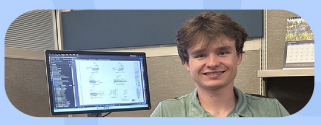
She sees a hub from 2024 titled 'Budgeting Mexico - Italy' and opens the hub. When previewing the hub, she can see a variety of pins and tabs titled VISA costs, accommodation, travel, and fees. When exploring, she can see the hub creator has placed pins along their timeline for every cost associated with their journey. When Marisol clicks the accommodation tab, she can preview all associated pins in a list, noting that this user paid a $500 deposit on the 3rd of August and $900 for their first month's rent on the 23rd.

She decides to save the hub and begins searching for others on the same topic. After saving several more hubs, she can directly compare them on her profile page. Marisol is emboldened by the detailed, community-generated content GSH provides, which helps her budget and plan her journey abroad. When she's ready, Marisol plans to create or join a public hub, hoping to find like-minded migrants from Mexico who are planning the same journey.

Daniyar Khusanov

Age: 22 | Nationality: Khazakhstani | Income: $2100 pm

Home: Astana | Job: Tech Intern | Priority: University Ranking



Daniyar is motivated to use GSH because of its in-depth planning and administration features. When he opens GSH, he's presented with two options: to create a hub or search for a hub. First, Daniyar uses the search function for inspiration, viewing archived hubs for Munich and Dublin. He's able to find previous communities made by users from his region, noting that the administrative burden for Munch is much lower.

After doing additional research, Daniyar decides the Technical University of Munich is a perfect fit, so he creates a public Study Hub for planning and administration, allowing users to request to join. He selects his home country and target country, choosing the 'request to join' privacy setting. He adds the tags 'community' and 'administration' to his hub to make it easier to find.

Daniyar receives three requests for his study hub, all from Kazakhstan. He is surprised to find peers who are making the same journey. They collaborate, creating tabs for their VISAs, accommodation planning, budgeting, travelling, and residency permits. The four members decide to rent an apartment together, coordinating their deposit, rent, and viewings using their Study Hub. Daniyar feels supported by GSH to build a community. When problems arise, he can ask his peers for guidance.

8.0 Development

8.1 Selecting our Final Concept

After completing both prototypes, we conducted a SWOT analysis to select our final concept. We felt GSH succeeded in community building, peer-to-peer support, and depth of functionality. However, we felt it had a high barrier to entry, was deeply complex, and focused too heavily on the planning stage. While we saw an opportunity for meaningful community support, we were also concerned about safety, privacy and content moderation.

For ICS, we felt it succeeded in accessibility and making comparisons. However, we felt its simplicity was restrictive, and it lacked social elements for community building. While we saw an opportunity to enhance decision-making, we were concerned that users might feel limited and isolated. Ultimately, we decided to bring forward Immigration Cheat Sheets because they addressed more of the requirements.

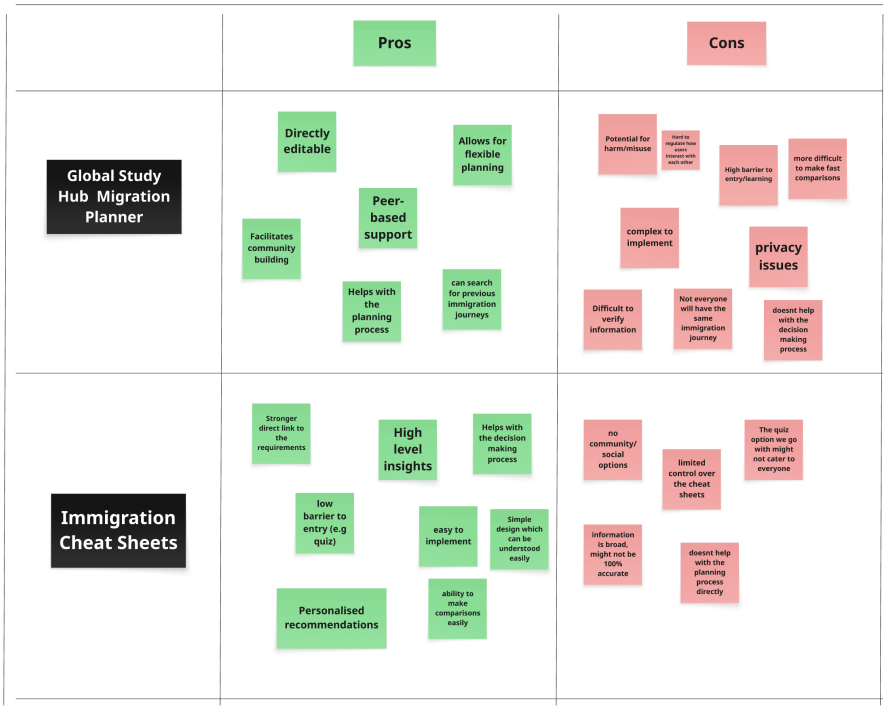


Figure 12: Comparing ICS to GSH

8.2 Hi-Fi Design

The creative design for ICS is rooted in a colourful 'folder' motif, which reflects the educational theme. We wanted the user to feel as if they were organising and planning manually, without being too heavy-handed. Each section is divided by colour to aid navigation. Interactive elements are enclosed in a minimalist 'button shape' to help guide the user. We opted for natural language buttons, avoiding graphic representations unless they were clear and easily understood. When icons are used, they are paired with titles that describe functionality to aid the user. The layouts maintain a top-down size hierarchy, guiding the user through the preferred order of viewing or interacting.

Figure 13: Hi-Fi Prototype illustrating

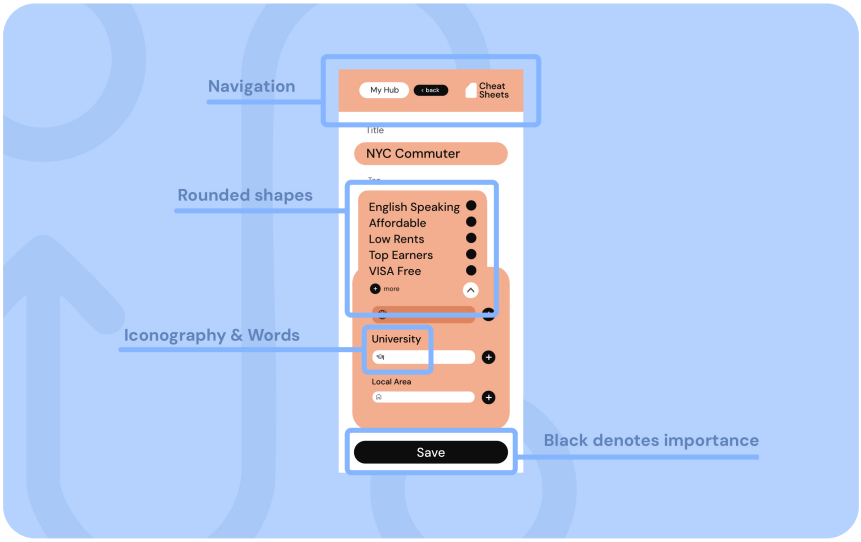


Figure 13 (a): Key UI Elements and Visual Design Decisions

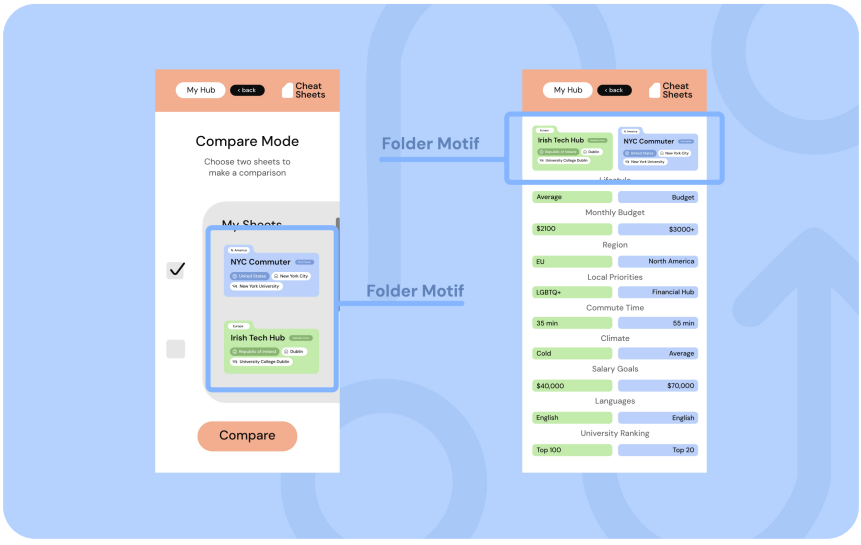


Figure 13(b): Comparison Flow and Folder-Based Motif

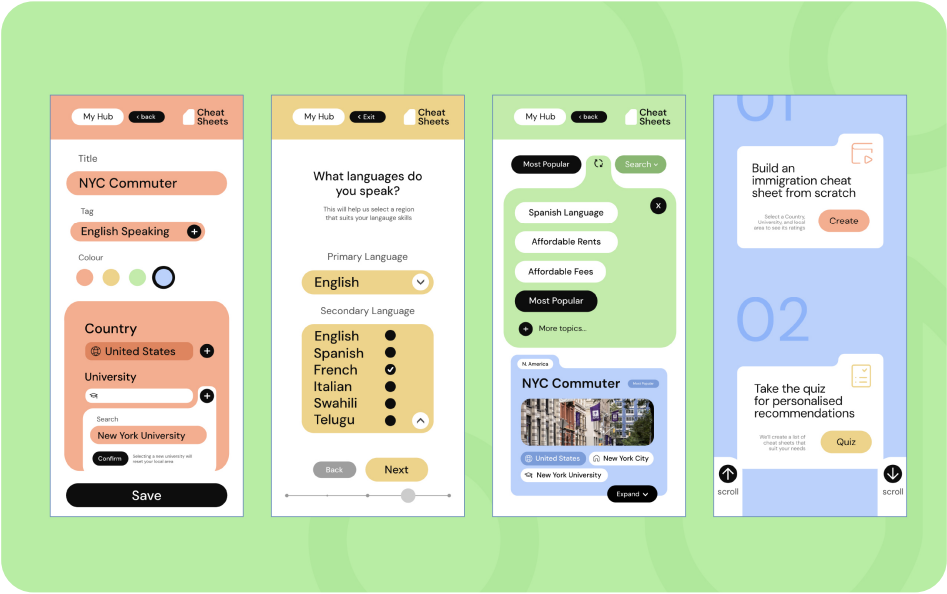


Figure 13(c): Final High-Fidelity Screens Across Core User Tasks

8.3 Hi-Fi Output

Because the user evaluation indicated that participants were unsure how to interact with the prototype, we refined the first page to include a clearer onboarding explanation. We also revised several quiz questions to provide more context and reduce ambiguity, ensuring that users understood what each question required.

https://patrickfeeneydesign.com/ICS.pptx

Quiz Pathway

The quiz comprises 10 questions, each designed to elicit a specific user preference regarding lifestyle, location, academic goals, and long-term planning.

Explore Pathway

The Explore module provides two navigation options. Users can either browse through predefined topics to view recommended content or use a search mode with filters to quickly find what they need without endless scrolling.

Create Pathway

The Plan module keeps the original features, creating, viewing, and comparing plans, but introduces colour-coded design elements to make plans easier to recognise and organise.

9.0 Evaluation

9.1 Results of Heuristics

For our prototype, a heuristic evaluation was carried out using Nielsen's usability heuristics. (Nielsen et al., 1990) Overall, this process highlighted recurring problems around unclear instructions, error prevention on the quiz, limited navigation options, inconsistent interaction patterns, and a lack of transparency around recommendations. The initial prototype lacked a central hub and confusing navigation. Furthermore, each section felt quite similar, leading to confusion about where users were inside the interface. The lack of early explanations led to confusion about the app's purpose. The quiz section also lacked a progress bar, so users didn't know how far along they were. Overall, insufficient attention was paid to the differences between casual and detailed use. For example, the explore page lacked quick links to popular searches, limiting users' options. (see Appendix G: Heuristic Evaluation)

9.2 Results of User Engagement

To ensure that our prototype aligned with real user needs, we conducted a user evaluation session with one of the participants from our research phase. By observing users interacting with our platform and gathering their feedback, we identified both strengths in the current design and areas requiring further improvement. The participant was given access to the high-fi prototype and asked to freely explore its features. Throughout the session, we encouraged them to think aloud so we could capture their expectations, moments of confusion, and overall reasoning. After the exploration session, the participant completed the System Usability Scale (SUS) questionnaire and provided open-ended feedback. This approach allowed us to combine structured usability scoring with qualitative insights into the user's natural interaction patterns. Note on SUS: because this evaluation involved a single participant and a free-exploration protocol (rather than a task-based test), SUS results are not statistically generalisable. We therefore treat the SUS response as a single, subjective indicator and place greater weight on the qualitative observations collected during the session.

What worked well

Visual design and icons: The participant praised the platform's overall visual style and the icons used throughout.

Simplicity and approachability: They described the interface as simple to use and easy to learn, which supported initial exploration without prior instruction.

Issues and opportunities

Ambiguous questions: Some quiz questions lacked necessary context (e.g., the “budget” question — monthly or annual?).

Missing education-level input: The quiz did not ask which level of education the user intended to pursue, affecting the relevance of the recommendations.

Results transparency: The university's recommendations were insufficiently detailed.

Missing Tutorial: Our platform should provide a guided tutorial for new users.

9.3 Changes Made

Quiz Clarity - We revised the language of our quiz to be easier to understand.

Navigation fixes - We implemented a central hub that aids navigation by providing easy access to the interface's sections.

Quick Links & Advanced Options - The explore page now includes popular tags for fast access to results, along with advanced search functionality, giving users more choices.

Detailed Results - We have added an option to expand on the recommendations provided, so our users can see more information about the results they are shown.

Quiz Overview - Users now see an overview of their answers at the end of the quiz before submitting, allowing them to spot errors before generating their results.

Quiz Progress - Users were given a progress bar for the quiz to provide greater contextual awareness, and a time estimate was also provided at the start.

Colour Coding - Introducing a central colour scheme for each section helped users more easily situate themselves within the system.

Improved onboarding - We introduced a home page with a clear explanation of the app's purpose, guiding the users through the different sections. However, we also included quick-links for more experienced users.

9.4 Future Improvements

Guided Tutorial - We plan to introduce a short first-time user walkthrough explaining the three main sections and how to interpret results.

Education Level - We plan to add a new question in our quiz section about the intended level of study.

Social Elements - While our prototype meets the requirements for comparing and unifying information, it doesn't address the need for collaboration and community building. A future iteration would introduce shared cheat sheets or folders and include social features such as commenting, direct messaging, and re-sharing.

Multi-Platform - Due to time constraints, our interface is available only on mobile devices at this stage. Future iterations should include more cross-platform functionality to enable more detailed use by students.